

KrackHack Hackathon Report

Offroad Semantic Scene Segmentation

Krackhack AI ML Track

Team Name: PS Codes

Team Members:

1. Saksham Setia

2.Prashant Suthar

Duality AI Offroad Autonomy Segmentation Challenge

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1 Problem Statement

The objective of this project is to develop a robust semantic segmentation model capable of accurately identifying terrain elements in desert environments. The task focuses on segmenting synthetic offroad scenes generated from digital twin simulations into predefined classes such as vegetation, terrain clutter, rocks, and sky.

Semantic segmentation is critical for autonomous offroad navigation because it enables unmanned ground vehicles to distinguish between traversable and non-traversable regions. Accurate segmentation helps in obstacle avoidance, path planning, and environment understanding in complex desert landscapes.

The dataset consists of synthetic images representing desert environments with pixel-level annotations. Models trained on one simulated environment must generalize to unseen desert environments, ensuring robustness against domain shifts.

2 Baseline

2.1 Overview

We implemented a baseline semantic segmentation model to establish a performance reference for the desert offroad dataset. The model segments synthetic desert scenes into predefined terrain classes and achieved a Mean Intersection over Union (mIoU) of **0.31** on the test set.

2.2 Model Architecture

A pretrained DINOv2 Vision Transformer was used as a frozen backbone to extract patch-level features. A lightweight ConvNeXt-style convolutional segmentation head was trained on top of these features to produce pixel-wise predictions.

2.3 Training Setup

The model was trained using cross-entropy loss with SGD optimizer for 10 epochs. Input images and masks were resized to match the backbone patch resolution. Training and validation were performed on separate splits to ensure fair evaluation.

2.4 Evaluation

Performance was measured using mIoU, Dice score, and pixel accuracy. The baseline results indicate moderate segmentation capability but reveal challenges in distinguishing

visually similar desert terrain classes.

3 About Dataset

The dataset comprises synthetic images generated from digital twin simulations of desert terrains. These images contain diverse environmental conditions including variations in lighting, vegetation density, and terrain structure.

Classes include desert-specific objects such as trees, bushes, dry grass, rocks, logs, and general landscape regions. Synthetic data enables controlled variation and large-scale annotation, which would otherwise be difficult to obtain in real desert environments.

Separate training, validation, and testing sets ensure fair evaluation on unseen scenes. Testing images represent new desert locations to assess model generalization.

3.1 Class Distribution Analysis

To better understand the dataset characteristics, we analyzed the pixel-wise class distribution for both training and testing sets.

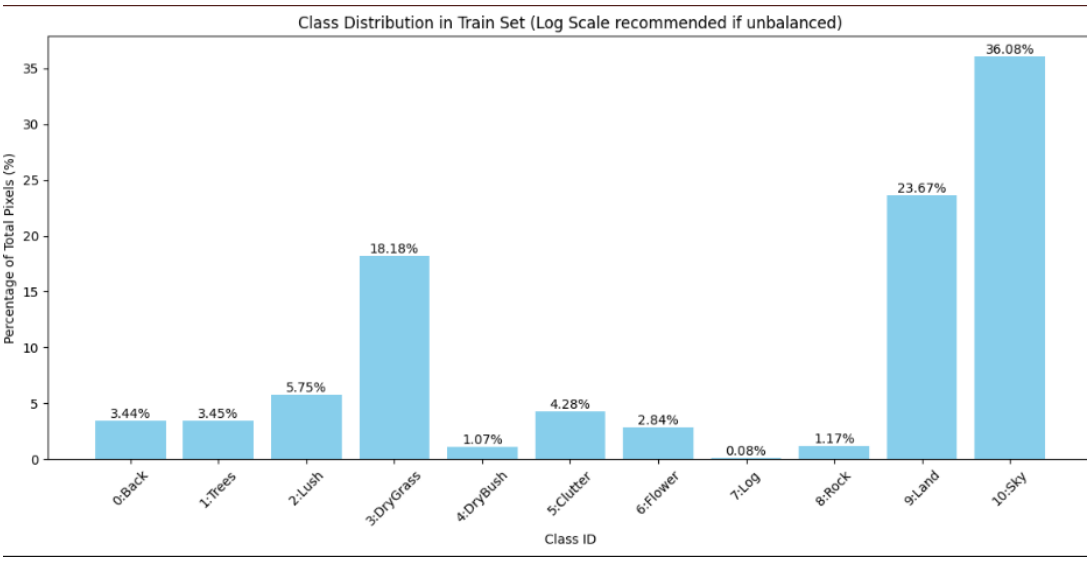


Figure 1: Class distribution in the training set

The training set shows significant class imbalance, with landscape and sky classes occupying the majority of pixels, while smaller object classes such as logs, flowers, and dry bushes are underrepresented.

The test set exhibits an even stronger imbalance and distribution shift compared to the training set. Some classes present during training appear with negligible or zero representation in the test set.

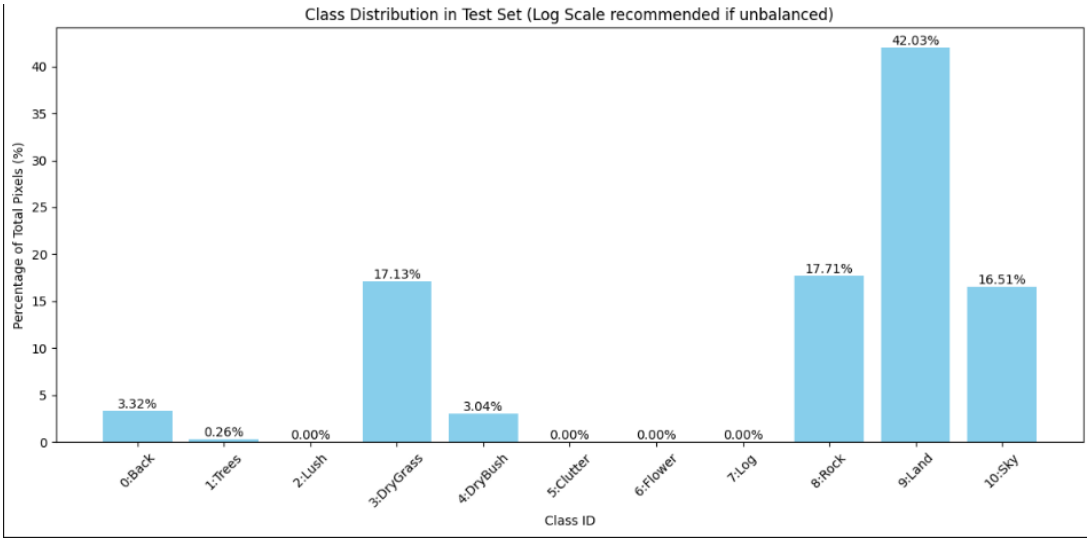


Figure 2: Class distribution in the test set

3.2 Challenges in Test Set Evaluation

This distribution mismatch introduces challenges in evaluating model performance on the test set:

- Certain classes are absent or extremely rare in the test data, making per-class metrics unstable.
- Models trained on dominant classes may achieve high overall accuracy while performing poorly on minority classes.
- Domain shift between training and testing environments affects generalization.

As a result, evaluation metrics such as IoU may not fully reflect the model's capability across all terrain categories, highlighting the importance of robust training strategies and balanced performance assessment.

4 Our Model

4.1 Data Augmentation

To improve generalization to unseen desert environments, a comprehensive augmentation pipeline was applied during training. Geometric transformations such as random horizontal flipping and affine transformations were used to simulate changes in camera viewpoint, vehicle motion, and terrain perspective. Vertical flipping was intentionally avoided to preserve realistic scene structure, as desert scenes always maintain consistent sky–ground orientation.

Photometric augmentations were applied to model environmental variability common in offroad settings. Adjustments in brightness, contrast, saturation, and hue simulated changes in lighting conditions, weather, and vegetation appearance, while occasional Gaussian blur represented motion blur and focus variations. These transformations were applied only to input images to maintain the correctness of segmentation masks.

Special care was taken to preserve semantic label integrity during preprocessing. Masks were resized using nearest-neighbor interpolation to avoid introducing invalid class values, and raw pixel labels were mapped to contiguous class indices for efficient training. This augmentation strategy exposed the model to diverse visual conditions and improved robustness against domain shifts between training and testing environments.

4.2 Model Architecture

To address the challenges of semantic segmentation in desert offroad environments, we designed a multi-scale encoder–decoder architecture that combines transformer-based semantic understanding with convolutional spatial refinement.

4.2.1 Encoder: Transformer-Based Feature Extraction

A pretrained DINOv2 Vision Transformer serves as the encoder. The backbone extracts hierarchical features representing different levels of scene understanding:

- Early layers capture low-level textures such as sand patterns and vegetation edges
- Intermediate layers capture object parts such as bushes, rocks, and logs
- Deep layers capture high-level semantic context such as terrain layout and sky regions

Multiple intermediate features are extracted from different transformer blocks to provide multi-scale information for the decoder.

4.2.2 Context Module: ASPP

An Atrous Spatial Pyramid Pooling (ASPP) module processes the deepest encoder features to capture context at multiple spatial scales. This is particularly important in desert environments where objects vary significantly in size, from small rocks to large terrain regions.

The ASPP module aggregates global and local context, enabling the model to distinguish visually similar terrain classes.

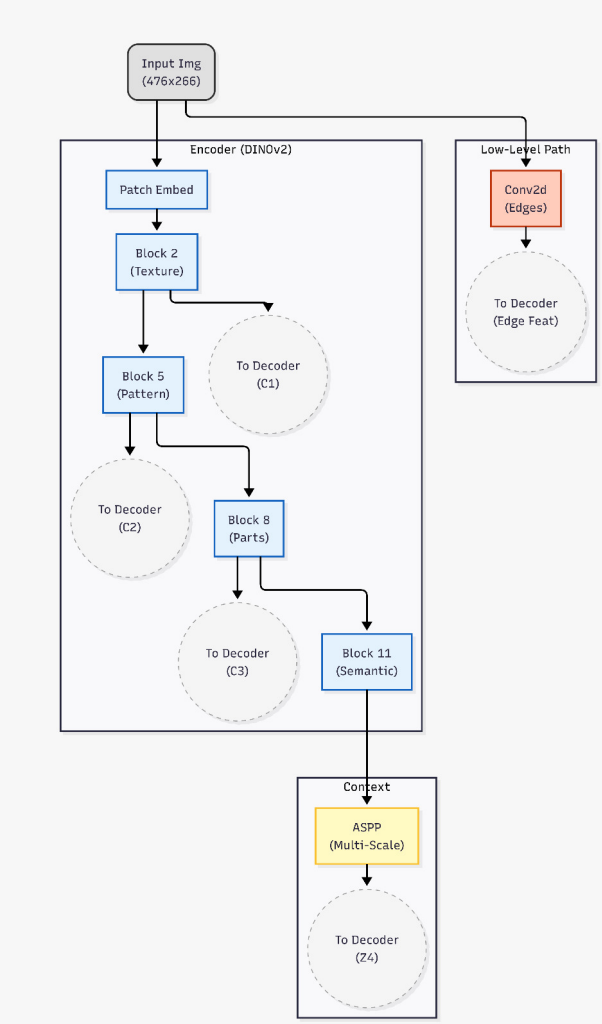


Figure 3: Transformer encoder with multi-scale feature extraction

4.2.3 Attention-Guided Decoder

The decoder progressively reconstructs high-resolution segmentation maps using upsampling and skip connections.

Attention gates are applied to encoder features before fusion, allowing the model to emphasize relevant terrain regions while suppressing background noise. This helps focus on meaningful structures such as vegetation clusters and obstacles.

Residual blocks are used at each decoding stage to refine features and maintain stable gradient flow during training.

4.2.4 Low-Level Feature Fusion

To recover fine spatial details lost during encoding, low-level features are extracted directly from the input image using convolutional layers. These features capture edges and texture information critical for accurate boundary prediction.

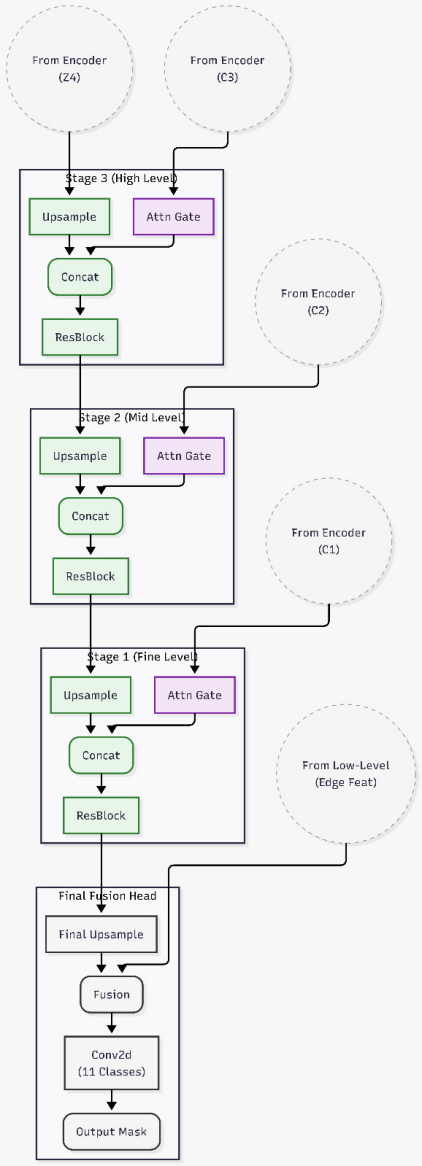


Figure 4: Attention-guided decoder with residual refinement

The final fusion stage combines deep semantic features with these low-level details to produce a high-resolution segmentation mask.

4.2.5 Output Head

The fused feature map is processed by convolutional layers to generate pixel-wise predictions for all terrain classes. The final output represents the segmented desert scene, distinguishing between vegetation, obstacles, terrain types, and sky.

This architecture effectively balances global context and local detail, making it suitable for complex offroad environments with diverse terrain structures.

4.3 Loss Function

Semantic segmentation in desert environments involves severe class imbalance and visually similar terrain categories. To address these challenges, we employed a hybrid loss function that combines Cross-Entropy Loss with Lovász-Softmax Loss.

4.3.1 Cross-Entropy Loss

Cross-Entropy Loss provides stable pixel-wise supervision and encourages correct class predictions for each pixel. It is effective for optimizing classification accuracy during training.

4.3.2 Lovász-Softmax Loss

While Cross-Entropy focuses on per-pixel accuracy, segmentation performance is evaluated using Intersection over Union (IoU). To directly optimize for IoU, we incorporated the Lovász-Softmax Loss, which approximates the gradient of the Jaccard index.

This loss improves boundary prediction and helps the model better segment irregular terrain structures such as bushes, rocks, and clutter.

4.3.3 Combined Loss Strategy

The final loss is a weighted combination:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \cdot \mathcal{L}_{Lovasz} \tag{1}$$

where $\lambda = 0.6$ controls the contribution of the Lovász term.

This combination balances stable optimization with direct improvement in IoU, leading to better performance on minority terrain classes and complex desert scenes.

The hybrid loss function proved effective in improving segmentation quality compared to using Cross-Entropy alone.

5 Result and Performance Metrics

The proposed model was evaluated on the unseen test dataset to assess its generalization capability to new desert environments. The model achieved a Mean Intersection over Union (mIoU) of **0.4603**, representing a **48.5% improvement** over the baseline mIoU of 0.31.

5.1 Per-Class IoU

Performance varied across terrain categories due to class imbalance and distribution differences between training and test sets.

- Trees : 0.6033
- Lush Bushes: 0.0007
- Dry Grass: 0.4929
- Dry Bushes: 0.4096
- Rocks8: 0.0514
- Landscape: 0.6764
- Sky: 0.9881

Some classes were absent in the test dataset, leading to undefined metrics for those categories.

5.2 Inference Performance

Inference time was measured on both GPU and CPU to evaluate deployment feasibility:

- GPU Runtime: 44 ms per image
- CPU Runtime: 347 ms per image

The results demonstrate that the model is capable of near real-time inference on GPU hardware, making it suitable for offroad autonomous systems.

5.2.1 Model Weights

The trained model weights are publicly available for reproducibility and further experimentation:

Download Model Weights

6 Challenges

- **Limited GPU Availability:** Training transformer-based segmentation models required significant computational resources. Due to restricted GPU access, experiments had to be conducted with small batch sizes and limited iterations.
- **Time Constraints:** The hackathon format imposed strict time limits, reducing the ability to perform extensive hyperparameter tuning and model experimentation.
- **Domain Shift Between Train and Test Data:** The test set contained desert scenes from unseen environments with different terrain patterns, vegetation density, and lighting conditions, making generalization challenging.
- **Class Imbalance:** Certain classes such as landscape and sky dominated the dataset, while smaller objects like logs and rocks were underrepresented, affecting model performance on minority classes.
- **Visual Similarity Between Classes:** Many terrain categories shared similar textures and colors, making accurate boundary prediction difficult.

7 Conclusion

This project presented a semantic segmentation approach tailored for offroad desert environments using synthetic digital twin data. The proposed pipeline integrated transformer-based feature extraction, multi-scale context aggregation, attention-guided decoding, and a hybrid loss function to enhance segmentation performance. The model demonstrated improved scene understanding compared to the baseline, particularly in complex terrain regions containing visually similar classes. Throughout the project, challenges such as domain shift between training and testing environments, class imbalance, limited computational resources, and strict time constraints were encountered. These findings highlight the difficulty of generalizing segmentation models to unseen offroad scenarios and emphasize the importance of robust training strategies. Future work may explore domain adaptation techniques, advanced augmentation methods, and validation on real-world datasets to further improve model reliability and applicability.